

German University in Cairo (GUC)
Media Engineering and Technology (MET)

Automated Hallucination Detection Report

Neuro-Symbolic Verification via Knowledge Graphs

Bachelor Thesis Project Output

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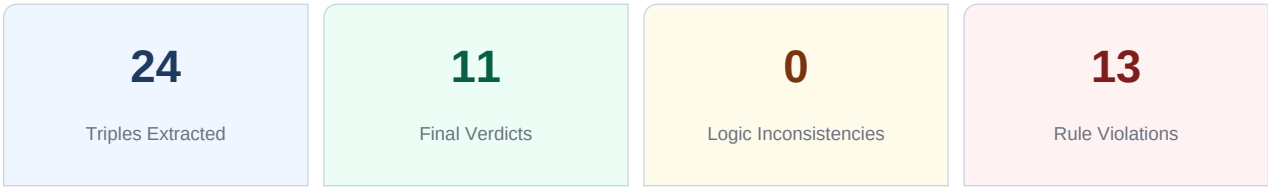
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REPORT OVERVIEW & METHODOLOGY

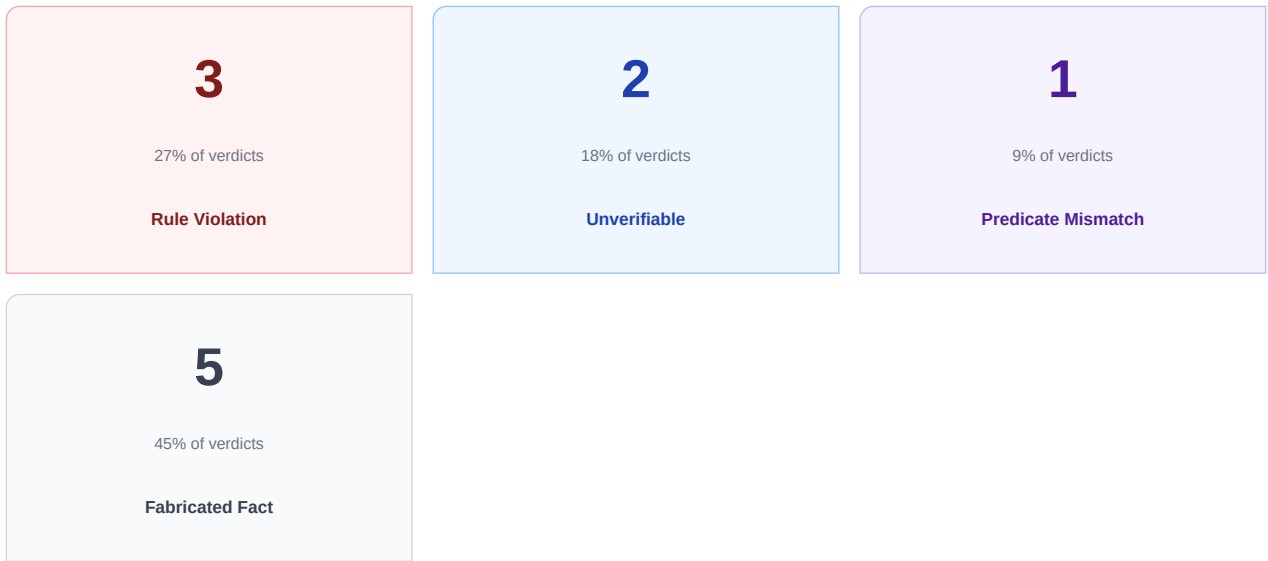
This document outlines the results of the KG-TripleGuard verification pipeline, a neuro-symbolic approach designed to enforce domain rules and verify factual consistency in Black-Box Large Language Models (LLMs).

Pipeline Stages:

1. **Triple Extraction & Canonicalization:** Unstructured LLM outputs are decomposed into relational triples (Subject, Predicate, Object) and mapped to the formal schema of the underlying database.
2. **Symbolic Constraint Verification:** Triples are subjected to rigorous logical constraint checks to identify domain-rule violations prior to graph traversal.
3. **Graph-based Factual Verification:** Canonicalized triples are queried against the Neo4j Knowledge Graph to categorize them as strictly correct, unverifiable, or explicitly fabricated.



VERDICT DISTRIBUTION



EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response prior to canonicalization or knowledge-graph checking.

#	Subject	Predicate	Object	Type	Predicate Definition
1	890	HAS_INTEREST_RATE	30%		The interest rate percentage associated with the subject.
2	290	HAS_INTEREST_RATE	20%		The interest rate percentage associated with the subject.
3	333	HAS_INTEREST_RATE	40%		The interest rate percentage associated with the subject.
4	333	HAS_RISK_TIER	high prime		The risk category or tier assigned to the subject.
5	598	HAS_RISK_TIER	high prime		The risk category or tier assigned to the subject.
6	633	HAS_RISK_TIER	high prime		The risk category or tier assigned to the subject.
7	729	HAS_RISK_TIER	high prime		The risk category or tier assigned to the subject.
8	833	HAS_RISK_TIER	high prime		The risk category or tier assigned to the subject.
9	890	HAS_RISK_TIER	high prime		The risk category or tier assigned to the subject.
10	290	HAS_RISK_TIER	high prime		The risk category or tier assigned to the subject.
11	301	HAS_RISK_TIER	deep subprime		The risk category or tier assigned to the subject.
12	333	HAS_APPROVAL_ODDS	200		The numerical value representing the odds of approval for the subject.
13	290	HAS_APPROVAL_ODDS	30		The numerical value representing the odds of approval for the subject.
14	890	HAS_APPROVAL_ODDS	40		The numerical value representing the odds of approval for the subject.
15	300	HAS_INTEREST_RATE	2%		The interest rate percentage associated with the subject.
16	303	HAS_INTEREST_RATE	3%		The interest rate percentage associated with the subject.
17	333	HAS_INTEREST_RATE	4%		The interest rate percentage associated with the subject.
18	777	HAS_INTEREST_RATE	6%		The interest rate percentage associated with the subject.
19	loan_100000_usd	REQUIRES_ACCOUNT_BALANCE	Not explicitly provided in the given context. Typically, lenders require proof of income, assets, or a certain debt-to-income ratio rather than a fixed account balance.		The required account balance for the subject loan.
20	300	HAS_DESCRIPTION	Deep subprime (very poor credit, high risk of default, limited access to credit)		The descriptive text explaining the subject's characteristics.
21	300	HAS_APPROVAL_ODDS	1		The numerical value representing the odds of approval for the subject.
22	305	HAS_APPROVAL_ODDS	2		The numerical value representing the odds of approval for the subject.
23	405	HAS_APPROVAL_ODDS	3		The numerical value representing the odds of approval for the subject.
24	730	HAS_APPROVAL_ODDS	4		The numerical value representing the odds of approval for the subject.

CANONICALIZED TRIPLES

Predicates normalized against the database schema. The Confidence column reflects alignment quality.

Subject	Original Predicate	Object	Canonical Predicate	Conf.
890	HAS_INTEREST_RATE	30%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
290	HAS_INTEREST_RATE	20%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_INTEREST_RATE	40%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
598	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
633	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
729	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
833	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
890	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
290	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
301	HAS_RISK_TIER	deep subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
333	HAS_APPROVAL_ODDS	200	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
290	HAS_APPROVAL_ODDS	30	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
890	HAS_APPROVAL_ODDS	40	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
300	HAS_INTEREST_RATE	2%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
303	HAS_INTEREST_RATE	3%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_INTEREST_RATE	4%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
777	HAS_INTEREST_RATE	6%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
loan_100000_usd	REQUIRES_ACCOUNT_BALANCE	Not explicitly provided in the given context. Typically, lenders require proof of income, assets, or a certain debt-to-income ratio rather than a fixed account balance.	REQUIRES_ACCOUNT_BALANCE	LOW

Subject	Original Predicate	Object	Canonical Predicate	Conf.
300	HAS_DESCRIPTION	Deep subprime (very poor credit, high risk of default, limited access to credit)	HAS_DESCRIPTION	LOW
300	HAS_APPROVAL_ODDS	1	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
305	HAS_APPROVAL_ODDS	2	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
405	HAS_APPROVAL_ODDS	3	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
730	HAS_APPROVAL_ODDS	4	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM

FACTUALLY INCONSISTENT TRIPLES

All triples labelled `factually_inconsistent` from the combined verification agent. Part A shows Pass-1 symbolic constraint violations. Part B shows Pass-2 hallucination verdicts with full evidence.

PART A — Symbolic Constraint Violations (Pass 1)

Label	Inconsistency Type	Subject	Predicate	Object	Constraint ID	Explanation
factually_inconsistent	RULE_VIOLATION	890	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	30%	c_interest_rate_range	The credit score of 890 is out of range because it is greater than 850.
factually_inconsistent	RULE_VIOLATION	290	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	20%	c_interest_rate_range	The credit score of 290 is out of range because it is less than 300.
factually_inconsistent	RULE_VIOLATION	333	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	40%	c_interest_rate_range	The interest rate of 40% is out of range because it is greater than 30.
factually_inconsistent	RULE_VIOLATION	333	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_300_579	The credit score of 333 is between 300 and 579, but the risk tier 'high prime' does not contain 'deep subprime'.
factually_inconsistent	RULE_VIOLATION	598	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_580_619	The credit score of 598 is between 580 and 619, but the risk tier 'high prime' does not contain 'subprime'.
factually_inconsistent	RULE_VIOLATION	633	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_620_659	The credit score of 633 is between 620 and 659, but the risk tier 'high prime' does not contain 'near prime'.
factually_inconsistent	RULE_VIOLATION	833	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_740_850	The credit score of 833 is between 740 and 850, but the risk tier 'high prime' does not contain 'super prime'.
factually_inconsistent	RULE_VIOLATION	890	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_740_850	The credit score of 890 is between 740 and 850, but the risk tier 'high prime' does not contain 'super prime'.
factually_inconsistent	RULE_VIOLATION	290	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_300_579	The credit score of 290 is between 300 and 579, but the risk tier 'high prime' does not contain 'deep subprime'.
factually_inconsistent	RULE_VIOLATION	333	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	200	c_approval_odds_range	The approval odds value of 200 is outside the valid range of 0-100.
factually_inconsistent	RULE_VIOLATION	290	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	30	c_approval_odds_range	The credit score of 290 is outside the valid range of 300-850.
factually_inconsistent	RULE_VIOLATION	890	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	40	c_approval_odds_range	The credit score of 890 is outside the valid range of 300-850.

Label	Inconsistency Type	Subject	Predicate	Object	Constraint ID	Explanation
factually_inconsistent	RULE_VIOLATION	405	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	3	c_approval_odds_range	The credit score 405 is within the valid range, but the approval odds 3 is less than 100, however, the credit score is valid and the approval odds is also valid, the issue is the credit score is less than 300 is not true, but the approval odds is less than 100 is true, however, the constraint explanation says a credit score outside 300-850 always violates, but in this case, the credit score is within the range of a valid credit score, but the explanation also says an approval odds value outside 0-100 always violates, and in this case, the approval odds is within the range of 0-100, so the issue is actually the credit score is less than the minimum valid credit score is not true, but the explanation says out-of-range subject is never an auto-pass, so the issue is actually the credit score is valid, but the explanation says a credit score outside 300-850 always violates, so the issue is the explanation itself, however, the explanation also says an approval odds value outside 0-100 always violates, and in this case, the approval odds is within the range of 0-100, so the issue is actually the explanation, but the explanation says the subject is less than 300 or greater than 850, and in this case, the subject is less than 300 is not true, because the subject is 405, which is greater than 300, so the issue is actually the explanation, so the correct explanation is the subject is within the valid range, and the approval odds is also within the valid range, so the correct classification is passed, but the explanation says a credit score outside 300-850 always violates, so the correct explanation is the subject is within the valid range, and the approval odds is also within the valid range, but the explanation says an approval odds value outside 0-100 always violates, so the correct explanation is the subject is within the valid range, and the approval odds is also within the valid range, so the correct classification is passed, but the explanation says out-of-range subject is never an auto-pass, so the correct explanation is the subject is within the valid range, and the approval odds is also within the valid range, so the correct classification is passed.

PART B — Hallucination Detection Verdicts (Pass 2)

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
729	RISK_TIER_CLASSIFIED_AS_Borrower_DANGER_LEVEL	high prime	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	SUBJECT	■■■■■■■ ■■■■ 0.00	The subject and object do not exist in the DB candidates.

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
301	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	deep subprime	Rule Violation	factually_inconsistent	RULE_VIOLATION	SEMANANTIC	SUBJECT	●●●●●●■ ■■■ 0.60	The subject does not match any DB candidate, but the predicate and object do. Ev: {"subject": "375", "predicate": "RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL", "object": "deep subprime"}
300	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	2%	Rule Violation	factually_inconsistent	RULE_VIOLATION	STRICT	OBJECT	●●●●●●● ●●● 1.00	The subject and predicate match a DB candidate, but the object does not. Ev: {"subject": "300", "predicate": "INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE", "object": "22.94"}
303	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	3%	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	SUBJECT	■■■■■■■ ■■■ 0.00	The subject and object do not exist in the DB candidates.
333	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	4%	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	SUBJECT	■■■■■■■ ■■■ 0.00	The subject and object do not exist in the DB candidates.
777	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	6%	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	SUBJECT	■■■■■■■ ■■■ 0.00	The subject and object do not exist in the DB candidates.
loan_100000_usd	REQUIRES_ACCOUNT_BALANCE	Not explicitly provided in the given context. Typically, lenders require proof of income, assets, or a certain debt-to-income ratio rather than a fixed account balance.	Unverifiable	unverifiable	—	NONE	—	■■■■■■■ ■■■ 0.00	The predicate is not known in the DB schema and there are no DB candidates.

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
300	HAS_DESCRIPTION	Deep subprime (very poor credit, high risk of default, limited access to credit)	Unverifiable	unverifiable	—	NONE	—	0.00	The predicate is not known in the DB schema and there are no DB candidates.
300	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	1	Rule Violation	factually_inconsistent	RULE_VIOLATION	STRIT	OBJECT	1.00	The subject and predicate match a DB candidate, but the object does not. Ev: {"subject": "300", "predicate": "APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD", "object": "8"}
305	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	2	Predicate Mismatch	factually_inconsistent	PREDICATE_MISMATCH	NONE	PREDICATE	0.60	The subject exists in the DB candidates, but the predicate does not match any known relation for the subject. Ev: {"subject": "305", "predicate": "RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL", "object": "deep subprime"}
730	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	4	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	SUBJECT+OBJECT	0.40	The subject exists in the database, but the predicate and object do not match any known relation for the subject. Ev: [{"subject": "730", "predicate": "RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL", "object": "prime"}, {"subject": "730", "predicate": "INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE", "object": "7.29"}]

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of any external knowledge source.

No logical inconsistencies detected.

VERDICT LEGEND

Verdict	Description
Correct	All three triple components verified against the knowledge graph.
Rule Violation	One or more components contradict database evidence.
Multi-Rule Violation	Violation spans multiple knowledge-graph triples simultaneously.
Unverifiable	No usable DB evidence found for any component of this triple.
Predicate Mismatch	Subject entity exists, but the stated predicate is not a valid relation.
Fabricated Fact	Triple asserts a relationship absent from the knowledge graph entirely.